Day_2 – VALUE TYPE CONVERSION SOLUTIONS

Question 1 :

```
Day_2_Exercise        Day_2_Exercise.Q1        Main(string[] args)
namespace Day_2_Exercise
{
    0 references
    internal class Q1
    {
        0 references
        static void Main(string[] args)
        {
            int totaldays = 100;
            int attended_days = 60;
            double percentage = (((double)attended_days/(double)totaldays)*100);
            Console.WriteLine("the percentage of student is : " + Convert.ToInt32(percentage)+"%");

            //Convert.ToInt32 is rounding and (int) is truncation which means in rounding if after decimal is greater than 0.5 then
            // it will round to the next number and in truncation it will stop or cuts the value after the decimal

        }
    }
}
```

Question 2:

```
Day_2_Exercise        Day_2_Exercise.Q2        Main(string[] args)
namespace Day_2_Exercise
{
    0 references
    internal class Q2
    {
        0 references
        static void Main(string[] args)
        {
            int maths = 50;
            int sst = 60;
            int eng = 70;
            decimal avg = (decimal)(maths + eng + sst) / 3.0m;

            Console.WriteLine("Avg : " + avg.ToString("F2")); // instead of this "Avg : {avg:F2}" can be used
            // this basically converts the decimal value to string and F2 means 2 floating points

            Console.WriteLine("The scholarship is : " + Convert.ToInt32(avg));

            //conversion flow is : int -> decimal -> string -> int

            /* there is no precision loss in this case however if the numbers are changed and while converting
            to integer the floating points will be cut off*/
        }

    }
}
```

Question 3:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q3
    {
        0 references
        static void Main(string[] args)
        {
            decimal fine_per_day = 15m;
            int overdue_days = 7;
            double total_fine = ((double)fine_per_day* overdue_days);

            Console.WriteLine("The total fine is : "+Convert.ToDecimal(total_fine));

            // here different types is used becuase the fine_per_day is not fixed it could be in decimals also like 15.5
            /* Conversions occurs in 2 types implicitly (done by the compiler) and explicitly(done by the user)
             in this case explicitly converting the decimal to double and then again double to decimal*/
        }
    }
}
```

Question 4:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q4
    {
        0 references
        static void Main(string[] args)
        {
            decimal balance = 48000.12m;
            float ir = 2.5f;
            decimal monthly_ir = (balance * Convert.ToDecimal(ir)) / 100m * 12m;
            // divide by 100 because to remove the percent sign in the ir as 2.5f means 2.5% and divided by 12 for month
            Console.WriteLine("the monthly interest is : "+monthly_ir);

            //implicit conversion may fail because of the data loss
        }
    }
}
```

Question 5:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q5
    {
        0 references
        static void Main(string[] args)
        {
            double total = 250.5;
            decimal tax = 25m;
            decimal discount = 50m;

            decimal total_pay = (decimal)total + tax - discount;
            Console.WriteLine(total_pay);
            //since totalcart amount was double so converted into decimal
            //By using decimal there is no precision risks
        }
    }
}
```

Question 6 :

Day_2_Exercise Day_2_Exercise.Q6

```
namespace Day_2_Exercise
{
    0 references
    internal class Q6
    {
        0 references
        static void Main(string[] args)
        {
            short temp_read_1 = 200;
            short temp_read_2 = 300;
            short temp_read_3 = 400;

            double c1 = temp_read_1 / 10.0;
            double c2 = temp_read_2 / 10.0;
            double c3 = temp_read_3 / 10.0;

            //the temprature is divided by 10 measure in Celcius
            double avg = (c1 + c2 + c3) / 3;
            Console.WriteLine(Convert.ToInt32(avg));

            //If the temperature sensor sends a value outside range then overflow occurs
        }
    }
}
```

Question 7 :

```
namespace Day_2_Exercise
{
    0 references
    internal class Q7
    {
        0 references
        static void Main(string[] args)
        {
            double final = 78.0;
            byte grade;
            if (final >= 90)
                grade = 5;
            else if (final >= 75)
                grade = 4;
            else if (final >= 60)
                grade = 3;
            else if (final >= 40)
                grade = 2;
            else
                grade = 1;
            Console.WriteLine("Score: " + final);
            Console.WriteLine("Grade: " + grade);

            // validation is basically checking the input variablae before running
        }
    }
}
```

Question 8:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q8
    {
        0 references
        static void Main(string[] args)
        {
            long usage = 9876543210;
            double mb = usage / 1024.0 / 1024.0;
            double gb = usage / 1024.0 / 1024.0 / 1024.0;

            long roundMB = (long)Math.Round(mb);
            long roundGB = (long)Math.Round(gb);
            //rounded the values in long because the usage was tracked as long
            Console.WriteLine("Bytes: " + usage);
            Console.WriteLine("MB: " + roundMB);
            Console.WriteLine("GB: " + roundGB);

            //implicit conversions are done by the compiler from smaller to larger datatype
            // another rounding method is Math.floor() rounds to floor value
        }
    }
}
```

Question 9:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q9
    {
        0 references
        static void Main(string[] args)
        {
            int count = 12;
            ushort max_capacity = 500;

            Convert.ToInt32(max_capacity);
            // converted the ushort to int for final reporting to remove loss
            // signed no. represent both + and - but unsigned only +
        }
    }
}
```

Question 10:

```
namespace Day_2_Exercise
{
    0 references
    internal class Q10
    {
        0 references
        static void Main(string[] args)
        {
            int salary = 5500;
            double allowance = 1100;
            double deduction = 850;
            decimal net_salary = Convert.ToDecimal(salary)+Convert.ToDecimal(allowance)-Convert.ToDecimal(deduction);
            Console.WriteLine(net_salary);

            //type conversion flow = int->decimal; double->decimal
            // chose decimal becuase there will be no risk of loss
        }

    }
}
```